Modèles stochastiques en finance
Séance 4
Evaluation par EDP. Modèles à plusieurs actifs.

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Programme

Pricing with a PDE

- A Feynman-Kac formula
- The Black & Scholes PDE
- Numerical pricing with the Black & Scholes PDE ($\longrightarrow$ TP Question IV)

A multi-asset financial market framework

- No arbitrage and completeness
- Correlation in multidimensional models
- Multidimensional PDE

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Markov property

Let $T > 0$ be a finite horizon. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space equipped with a filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$.

Definition 1.1 (Markov property)

A stochastic process $(X_t)_{t \geq 0}$ is called a Markov process if for any $0 \leq s \leq t$ and for any bounded measurable function $f : \mathbb{R} \to \mathbb{R}$:

$$\mathbf{E}[f(X_t)|\mathcal{F}_s] = \mathbf{E}[f(X_t)|\sigma(X_s)].$$

- Future behavior of a Markov process does not depend on the past, but only on the observation at the current time.
- Alternative formulation: $\mathbf{E}[f(X_t)|\mathcal{F}_s] = g(s, X_s)$ for some measurable function $g : [0, T] \times \mathbb{R} \to \mathbb{R}$.

Itō diffusion

Let $(B_t)_{0\leq t\leq T}$ be a Brownian motion with natural filtration $(\mathcal{F}_t)_{0\leq t\leq T}$, and consider an Itō process solution of the stochastic differential equation

$$dX_t = a(t, X_t)\,dt + b(t, X_t)\,dB_t, \quad X_0 = x_0,$$

where the functions $a : [0, T] \times \mathbb{R} \to \mathbb{R}$ and $b : [0, T] \times \mathbb{R} \to \mathbb{R}$ satisfy the usual conditions, and $x_0 \in \mathbb{R}$.

- $a(t, X_t)$ is called the drift coefficient.
- $b(t, X_t)$ is called the diffusion coefficient.

The time-homogeneous case is written:

$$dX_t = a(X_t)\,dt + b(X_t)\,dB_t.$$

Markov property of Itō diffusions

For any $t \in [0, T]$ and $x \in \mathbb{R}$, let $X^{t,x} = (X_u^{t,x})_{u \in [t,T]}$ be an Itō diffusion on the time interval $[t, T]$ with the initial condition $X_t^{t,x} = x$. For simplicity, we set $X^x \triangleq X^{0,x}$.

Theorem 1.2 (Markov property of Itō diffusions)

Let $h : \mathbb{R} \to \mathbb{R}$ be a bounded measurable function. Then for any $0 \leq s \leq t \leq T$:

$$\mathbf{E}\left[h(X_t^x)|\mathcal{F}_s\right] = \varphi(s, X_s^x), \text{ where } \varphi(s,z) = \mathbf{E}[h(X_t^{s,z})].$$

(i.e. expliciting $\omega \in \Omega$: $\mathbf{E}\left[h(X_t^x)|\mathcal{F}_s\right](\omega) = \mathbf{E}[h(X_t^{s,X_s^x(\omega)})]$).

- Other frequently used notation: if $\mathbf{E}^x[h(X_t)]$ denotes the expectation computed using the distribution of X_t given $X_0 = x$, then the Markov property for Itō diffusion may be written

$$\mathbf{E}^x\left[h(X_{t+u})|\mathcal{F}_t\right] = \mathbf{E}^{X_t}[h(X_u)].$$

- Itō diffusions are Markov processes.

Conditional expectation and PDE

Theorem 1.3 (Conditional expectation and PDE)

Let X be an Itō diffusion on $[0, T]$ with drift $a(t, x)$ and diffusion coefficient $b(t, x)$. Let h be a measurable function. Let $g(t, x) = \mathbf{E}[h(X_T^{t,x})]$ and assume that this quantity is finite $\mathbf{P}$-a.s. Then g is a solution to the partial differential equation:

$$\frac{\partial g}{\partial t}(t, x) + a(t, x)\frac{\partial g}{\partial x}(t, x) + \frac{1}{2}b^2(t, x)\frac{\partial^2 g}{\partial x^2}(t, x) = 0,$$

with terminal condition: $\forall x \in \mathbb{R}, g(T, x) = h(x)$.

- Practical method: $g(t, X_t)$ is a martingale, compute $dg(t, X_t)$, set the drift to zero.

Some more vocabulary I

- The (infinitesimal) generator of an Itō diffusion is the functional operator A defined by

$$Af(x) = \lim_{t \to 0} \frac{\mathbf{E}[f(X_t^x)] - f(x)}{t},$$

when the limit exists for f and x.

- In the time-homogeneous case, for a function f nice enough (e.g., f twice differentiable with compact support)

$$Af(x) = a(x)\frac{\partial f}{\partial x}(x) + \frac{1}{2}b^2(x)\frac{\partial^2 f}{\partial x^2}(x),$$

and $f(X_t^x) - \int_0^t Af(X_s^x)\,ds$ is a $\mathcal{F}_t$-martingale.

Some more vocabulary II

- In the non-homogeneous case, if $(X_u^{t,x})_{u\in[t,T]}$ satisfies $dX_u = a(u, X_u)\,du + b(u, X_u)\,dB_u$, with $u \in [t, T]$ and initial state $X_t^{t,x} = x$, then $f(u, X_u^{t,x}) - \int_t^u \left(\frac{\partial f}{\partial u} + Af\right)(s, X_s^{t,x})\,ds$ is a $\mathcal{F}_u$-martingale, where

$$Af(t,x) = a(t,x)\frac{\partial f}{\partial x}(t,x) + \frac{1}{2}b^2(t,x)\frac{\partial^2 f}{\partial x^2}(t,x).$$

- We may then rewrite the first version of the Feynman-Kac theorem: if g is a solution of the partial differential equation

$$\left(\frac{\partial g}{\partial t} + Ag\right)(t,x) = 0, \quad t \in [0, T], x \in \mathbb{R},$$

with terminal condition $g(T, x) = f(x)$, then $g(t, x) = \mathbf{E}[f(X_T^{t,x})]$.

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The Black & Scholes PDE

Proposition 1.4 (The Black & Scholes PDE)

Let us consider a derivative product on the underlying asset S, with maturity T and payoff $H = h(S_T)$, with $h : \mathbb{R}_+ \to \mathbb{R}_+$. In the Black & Scholes model, the price $v(t, x)$ of this derivative product at time $t \in [0, T]$ given $S_t = x$ satisfies the following partial differential equation:

$$\frac{\partial v}{\partial t}(t, x) + rx\frac{\partial v}{\partial x}(t, x) + \frac{1}{2}\sigma^2 x^2 \frac{\partial^2 v}{\partial x^2}(t, x) = rv(t, x), \quad t \in [0, T], x \in \mathbb{R}_+,$$

with terminal condition: $v(T, x) = h(x)$, $x \in \mathbb{R}_+$.

- The Black & Scholes PDE does not depend on the (European) derivative product. The payoff impacts only the terminal condition.

The Black & Scholes formula for vanilla call options

- The Black & Scholes formula can be directly computed from the Black & Scholes PDE via the known solution of the homogeneous heat equation.

Proposition 1.5 (Black & Scholes formula for vanilla call options)

In the Black & Scholes model, the price C_t at time $t \in [0, T]$ of a European call option on the underlying S with maturity T and exercise price (strike) K is:

$$C_t = S_t \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

where Φ is the standard Gaussian cdf and:

$$d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, d_2 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}.$$

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Différences finies pour l'EDP de Black & Scholes I

- Renversement du temps : $\tau = T - t \longrightarrow$ condition initiale au lieu de terminale: $v(\tau, x) = p(T - \tau, x)$ est solution de l'équation aux dérivées partielles

$$\frac{\partial v}{\partial \tau}(\tau, x) - \frac{1}{2}\sigma^2 x^2 \frac{\partial^2 v}{\partial x^2}(\tau, x) - rx\frac{\partial v}{\partial x}(\tau, x) + rv(\tau, x) = 0,$$

  muni de la condition *initiale* $v(0, x) = p(T, x) = (x - K)_+$.
- Passage aux prix logarithmiques ($y = \log x$) $\longrightarrow$ EDP à coefficients constants: $u : [0, T] \times \mathbb{R} \to \mathbb{R}, (\tau, y) \mapsto u(\tau, y) = p(T - \tau, e^y)$ est solution de l'équation aux dérivées partielles :

$$\frac{\partial u}{\partial \tau}(\tau, y) - \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial y^2}(\tau, y) - \left(r - \frac{\sigma^2}{2}\right)\frac{\partial u}{\partial y}(\tau, y) + ru(t, y) = 0,$$

  muni de la condition initiale $u(0, y) = (e^y - K)_+$.

Différences finies pour l'EDP de Black & Scholes II

- Discrétisation en espace sur $[-L, L]$ avec $L >> \log K$: pour N entier, on pose $h = \frac{2L}{N}$, $y_i = -L + ih, i = 0, \ldots, N$
- Les valeurs $u(\tau, y_i)$ recherchées sont approchées par le vecteur $U(\tau) = (u_i(\tau))_{1 \leq i \leq N-1}$ tel que pour $i = 1, \ldots, N-1$:

$$\frac{du_i}{d\tau}(\tau) - \frac{\sigma^2}{2}\frac{u_{i+1}(\tau) - 2u_i(\tau) + u_{i-1}(\tau)}{h^2} - \left(r - \frac{\sigma^2}{2}\right)\frac{u_{i+1}(\tau) - u_{i-1}(\tau)}{2h} + ru_i(\tau) = 0.$$

- La discrétisation sur $[-L, L]$ suppose d'imposer des conditions aux bords.
  - Conditions de type Dirichlet : $u(\tau, -L) = \text{cste}$, $u(\tau, L) = \text{cste}$
  - Conditions de type Von Neumann : $\frac{\partial u}{\partial x}(\tau, -L) = \text{cste}$, $\frac{\partial u}{\partial x}(\tau, L) = \text{cste}$

Différences finies pour l'EDP de Black & Scholes III

- Sous forme matricielle, $\frac{dU}{d\tau}(\tau) = \mathbf{A}U(\tau) + B$, avec $\mathbf{A}$ matrice tridiagonale et B vecteur (suivant le choix des conditions aux bords):

$$\mathbf{A} = \begin{pmatrix} b & c & & & & \\ a & b & c & & & \\ & \ddots & \ddots & \ddots & & \\ & & a & b & c & \\ & & & a & b & \end{pmatrix}$$

- Dans le cas du call européen, les caractéristiques de l'option et la parité call-put permettent d'écrire pour tout $\tau \in]0, T[$:

$$\lim_{y \to -\infty} u(\tau, y) = 0, \text{ et } u(\tau, y) \underset{y \to +\infty}{\sim} e^y - Ke^{-r\tau}.$$

d'où les conditions de Dirichlet pour pricer un call européen par EDP:

$$u_0(\tau) = 0, \quad u_N(\tau) = e^{y_N} - Ke^{-r\tau}.$$

Différences finies pour l'EDP de Black & Scholes IV

- On part de l'équation $\frac{dU}{d\tau}(\tau) = \mathbf{A}U(\tau)$ et on discrétise $[0, T]$ en M pas de temps ($\Delta T = \frac{T}{M}$). Pour $j = 0, \ldots, M$, on approxime $(u_i(j\Delta T))_{1 \leq i \leq N-1}$ par le vecteur $U^j = (u_i^j)_{1 \leq i \leq N-1}$. U^0 est donné par la condition initiale.
- Schéma d'Euler explicite :
$$U^{j+1} = (\mathbf{I} + \Delta T \mathbf{A})U^j$$
- Schéma d'Euler implicite :
$$(\mathbf{I} - \Delta T \mathbf{A})U^{j+1} = U^j$$
- θ-schéma (Crank-Nicholson $\theta = \frac{1}{2}$) :
$$(\mathbf{I} - \theta \Delta T \mathbf{A})U^{j+1} = (\mathbf{I} + (1-\theta)\Delta T \mathbf{A})U^j$$

Différences finies pour l'EDP de Black & Scholes V

- Les schémas d'Euler implicite et de Crank-Nicholson (et de façon générale les θ-schémas pour $\theta \in \left[\frac{1}{2}, 1\right]$) sont (inconditionnellement) convergents.
- Le schéma d'Euler explicite (et de façon générale les θ-schémas pour $\theta \in \left[0, \frac{1}{2}\right[$) ne converge que sous condition (du type $\Delta T/h^2 \leq C$, i.e. fort accroissement du pas de temps, et donc du temps de calcul lorsqu'on affine le pas d'espace...).
- Ordre de convergence des schémas: 1 en temps et 2 en espace pour les schéma d'Euler ; 2 en temps et 2 en espace pour le schéma de Crank-Nicholson.
- La matrice $\mathbf{I} - \Delta T \mathbf{A}$ est tridiagonale : la résolution du schéma implicite ne nécessite pas d'inversion numérique de matrices, mais la solution est efficacement calculée par exemple par l'algorithme de Thomas.
- Une référence: "Computational methods for option pricing", Achdou, Pironneau, SIAM, 2005.

Pricing with a PDE, pros and cons

- Computational time: can be much faster than a Monte Carlo simulation (vector of prices, interpolation/adapted grid needed but greeks easily available)
- Flexibility: easily used for European payoffs (some tweaks may be need for numerical instabilities in the case of non-smooth payoffs ; e.g., greeks in the case of an up-and-out call when $S \uparrow B$ et $t \uparrow T$).
- Scope: Pricing PDEs can be obtained for some exotic options (asian option, lookback options) in the Black & Scholes model. Pricing PDEs can be obtained with more general asset models (e.g., with stochastic volatility, see next lecture).
- Curse of dimensionality: standard finite- difference schemes can't be used on a large number number of assets. High-dimensional schemes (splitting, sparse grids) not really satisfying.
- Recent advances: deep learning method may make PDE pricing more appealing (see next slide).

Numerical solutions of PDEs via learning

- General principle of the "deep Galerkin" method for a PDE on a function $u : [0, T] \times \Omega$:

$$\begin{cases} \frac{\partial u}{\partial \tau}(\tau, y) + \mathcal{A}u(\tau, y) = 0, & \text{(PDE, e.g. Black \& Scholes)} \\ u(0, x) = h(x) & \text{(initial condition)} \\ u(\tau, x) = v(\tau, x), \ x \in \partial\Omega & \text{(boundary conditions).} \end{cases}$$

- Idea : build a neural network with parameters θ that learns a function $u(\tau, y; \theta)$ by minimizing a loss function

$$\mathcal{L}(f) = \left\| \frac{\partial u}{\partial \tau}(\tau, y) + \mathcal{A}u(\tau, y) \right\|_2 + \|u(0, x) - h(x)\|_2 + \|u(t, x) - v(t, x)\|_2$$

  on some training sample (τ_n, y_n) in $[0, T] \times \Omega$.
- Open questions : performances ? high dimension ? sampling of the training dataset ? norms used in the loss function ? penalization ? hyperparameters ? etc.

TP Question IV.1

1. Tracer le prix EDP du call européen dans le modèle de Black & Scholes en fonction de S_0, puis tracer l'écart entre le prix EDP et le prix exact de la formule de Black & Scholes en fonction de S_0.
2. Tracer l'erreur en norme $\|\cdot\|_\infty$ du vecteur de prix EDP en fonction de N (à M suffisamment grand fixé). Retrouve-t-on l'ordre de convergence attendu ? Interpréter la forme de la courbe.
3. Tracer l'erreur en norme $\|\cdot\|_\infty$ du vecteur de prix EDP en fonction de M (à N suffisamment grand fixé). Retrouve-t-on l'ordre de convergence attendu ? Interpréter la forme de la courbe.
4. Tracer l'erreur ponctuelle $|u_i^j - u(j\Delta T, y_i)|$ en fonction de $t = T - j\Delta T \in [0, T]$ et $S = e^{y_i} \in [e^{-L}, e^L]$.

TP Question IV.2 (facultatif)

1. Réaliser (adapter, citer vos sources !) une implémentation Python (pytorch de préférence) de la méthode dite "Deep Galerkin" pour l'EDP de Black & Scholes. Bien expliquer l'architecture et la loss.
2. Tracer l'erreur en norme $\|\cdot\|_\infty$ du vecteur de prix EDP. Commenter.
3. Tracer l'erreur ponctuelle $|u_i^j - u(j\Delta T, y_i)|$ en fonction de $t = T - j\Delta T \in [0, T]$ et $S = e^{y_i} \in [e^{-L}, e^L]$.
4. Questions ouvertes:
   - Analyser la convergence de la méthode.
   - Tester la sensibilité aux hyperparamètres.
   - Tester la faisabilité en grande dimension (voir Section suivante).
   - ...

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Model description

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be some probability space. Let T be a finite horizon. Let $B = (B_1, \ldots, B_d)$ be a d-dimensional Brownian motion, and $(\mathcal{F}_t)_{t\in[0,T]}$ be its natural filtration. We consider a general model with m risky assets traded on the exchange. Let $S = (S_1, \ldots, S_m)$ be the vector of the risky assets prices. We assume that S_i, $i = 1, \ldots, m$ satisfies the stochastic differential equation

$$dS_i(t) = \mu_i(t)S_i(t)\,dt + S_i(t)\sum_{j=1}^{d}\sigma_{ij}(t)\,dB_j(t),$$

where $\mu_i, i = 1, \ldots, m$ and $\sigma_{ij}, i = 1, \ldots, m, j = 1, \ldots, d$ are $\mathcal{F}_t$-adapted processes.
As usual, we also assume that there exists a risk-free asset with price D satisfying the stochastic differential equation

$$dD_t = D_t r_t\,dt, \quad D_0 = 1,$$

where $(r_t)_{t\in[0,T]}$ is a $\mathcal{F}_t$-adapted process.

d-dimensional Girsanov theorem

Theorem 1.6 (A multidimensional version of the Girsanov theorem)

Let $\theta = (\theta_1, \dots, \theta_d)$ be an $\mathcal{F}_t$-adapted process s.t. $\int_0^T \|\theta(s)\|^2 \, ds < \infty$ $\mathbf{P}$-a.s.. For any $t \in [0, T]$, we set:

$$Z_t = \exp\left(-\int_0^t \theta(s) \cdot dB(s) - \frac{1}{2}\int_0^t \|\theta(s)\|^2 \, ds\right)$$

and

$$B^{\mathbf{Q}}(t) = B(t) + \int_0^t \theta(s) \, ds.$$

If Z is a martingale w.r.t. $(\mathcal{F}_t)$ and $\mathbf{P}$, then $(B_t^{\mathbf{Q}})_{0 \le t \le T}$ is a d-dimensional Brownian motion w.r.t. the filtration $(\mathcal{F}_t)_{0 \le t \le T}$ and the measure $\mathbf{Q}$ defined with $\frac{d\mathbf{Q}}{d\mathbf{P}} = Z_T$.

- Vector notations : $\int_0^t \theta(s) \cdot dB(s) = \sum_{j=1}^d \int_0^t \theta_j(s) dB_j(s)$ and $\int_0^t \theta(s) \, ds = \left(\int_0^t \theta_j(s) \, ds\right)_{j=1,\dots,d}$.
- Recall that Z is martingale if e.g., $\mathbf{E}\left[\int_0^T \|\theta(s)\|^2 Z_s^2 \, ds\right] < \infty$.

d-dimensional martingale representation theorem

Theorem 1.7 (d-dimensional martingale representation theorem)

If $(M_t)_{t\in[0,T]}$ is a square-integrable $\mathcal{F}_t$-martingale, then there exists a d-dimensional $\mathcal{F}_t$-adapted process $(\alpha(t))_{t\in[0,T]}$ satisfying $\mathbf{E}\left[\int_0^T \|\alpha(s)\|^2\, ds\right] < \infty$, and such that

$$\begin{aligned} M_t &= M_0 + \int_0^t \alpha_s \cdot dB_s \\ &= M_0 + \sum_{i=1}^{d} \int_0^t \alpha_i(s)\, dB_i(s) \quad \mathbf{P}-a.s., \quad t \in [0,T]. \end{aligned}$$

A multi-asset version of the first fundamental theorem

Proposition 1.8 (Martingale measure in the multi-asset case)

With sufficient technical conditions, if there exists a $\mathcal{F}_t$-adapted process $(\theta_t)_{t\in[0,T]}$ such that

$$\forall i = 1, \dots, m,\ \mu_i(t) - r(t) = \sum_{j=1}^{d} \sigma_{i,j}(t)\theta_j(t),$$

then the model admits an equivalent martingale measure, and there is no arbitrage.

- Consider a model with constant parameters, $m = 1$ and $d = 1$ (i.e. the Black & Scholes model).
- Consider a model with constant parameters, $m = 2$ and $d = 1$.
- Consider a model with constant parameters, $m = 1$ and $d = 2$.

A multi-asset version of the second fundamental theorem

Theorem 1.9 (Completeness in the multi-asset case)

Let us consider a general multi-asset financial market model such that there exists an equivalent martingale measure. The model is complete if, and only if, the equivalent martingale measure is unique.

- Key step : for a $\mathcal{F}_T$-measurable payoff H, if Γ is a process such that $d\left(\mathbf{E}^{\mathbf{Q}}\left[D_T^{-1}H \mid \mathcal{F}_T\right]\right) = \Gamma(s) \cdot dB^{\mathbf{Q}}(t)$, then the payoff can be replicated if we can find a process Δ such that

$$\forall j = 1, \ldots, d,\ \Gamma_j(t) = \sum_{i=1}^{m} \Delta_i(t)\tilde{S}_i(t)\sigma_{ij}(t).$$

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Correlated Brownian motions

- In the general model, the prices of m assets are determined by the same d coordinates $(B_1, \ldots, B_d)$ of a d-dimensional Brownian motion, which creates a correlation structure between the m assets.

Proposition 1.10 (Correlated Brownian motions)

Let $\sigma_i(t) = \sqrt{\sum_{j=1}^d \sigma_{ij}^2(t)}$. *For all* $i = 1, \ldots, m$, *assume that* σ_i *is positive* $\mathbf{P}$*-a.s., and define*

$$W_i(t) = \sum_{j=1}^d \int_0^t \frac{\sigma_{ij}(s)}{\sigma_i(s)}\, dB_j(s).$$

For any $i = 1, \ldots, m$, $(W_i(t))_{t\in[0,T]}$ *is a* $\mathcal{F}_t$*-Brownian motion.*

- The multi-asset model can thus be written

$$dS_i(t) = \mu_i(t) S_i(t)\, dt + \sigma_i(t) S_i(t)\, dW_i(t)\ ,$$

where $W_i, i = 1, \ldots, m$ are univariate Brownian motions that are *not necessarily independent*.

Instantaneous correlation

Definition 1.11 (Instantaneous correlation)

The $\mathcal{F}_t$-adapted process $\rho_{ij}(t)$ defined by

$$\rho_{ij}(t) = \sum_{k=1}^{d} \frac{\sigma_{ik}(t)\sigma_{jk}(t)}{\sigma_i(t)\sigma_j(t)},$$

is called instantaneous correlation of the Brownian motions W_i and W_j.

- $\rho_{ij}(t)$ can be seen as the linear correlation coefficient of the infinitesimal increments of the two Brownian motions at time t.

Simulating correlated Brownian motions

- A multi-asset model with constant correlation can be written
$$dS_i(t) = \mu_i(t)S_i(t)\,dt + \sigma_i(t)S_i(t)\,dW_i(t)\ ,$$
where $W_i, i = 1, \ldots, m$ are univariate Brownian motions correlated with correlation matrix $\boldsymbol{\rho} = (\rho_{ij})_{i,j}$ such that $d\langle W_i, W_j\rangle_t = \rho_{ij}dt$

Proposition 1.12 (Correlating Brownian motions)

Let B be a m-dimensional Brownian motion. Let $\boldsymbol{\rho}$ be a constant correlation matrix. There exists a lower triangular matrix $\mathbf{L}$ such that the coordinates of the process $W = \mathbf{L}B$ are univariate Brownian motions with correlation matrix $\boldsymbol{\rho}$.

- In the case $m = 2$, $\boldsymbol{\rho} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ gives $\mathbf{L} = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}$.
- This allows for easy Monte Carlo simulation of correlated Brownian motions, and thus correlated asset prices, i.e. correlated asset returns.

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Basket option on two underlying assets

- Two-dimensional Black & Scholes model:

$$\begin{cases} dS_t^1 = & rS_t^1\,dt + \sigma_1 S_t^1\,dW_t^1 \\ dS_t^2 = & rS_t^2\,dt + \sigma_2 S_t^2\,dW_t^2 \end{cases}$$

  with correlation $d\langle W^1, W^2\rangle_t = \rho\,dt$.
- Basket payoff $h(S_T^1, S_T^2)$
- Then the price $p : [0, T] \times (\mathbb{R}_+)^2 \to \mathbb{R}_+, (t, x, y) \mapsto p(t, x, y)$ satisfies

$$\frac{\partial p}{\partial t}(t,x,y) + \frac{1}{2}\sigma_1^2 x^2 \frac{\partial^2 p}{\partial x^2}(t,x,y) + \rho\sigma_1\sigma_2 xy \frac{\partial^2 p}{\partial x \partial y}(t,x,y) + \frac{1}{2}\sigma_2^2 x^2 \frac{\partial^2 p}{\partial y^2}(t,x,y)$$
$$+ rx\frac{\partial p}{\partial x}(t,x,y) + ry\frac{\partial p}{\partial y}(t,x,y) = rp(t,x,y),$$

  with terminal condition $p(T, x, y) = h(x, y)$.